Lecture 13

The characters of S_n

Objectives. *Lecture 12 labelled the irreducibles of S_n by partitions. This lecture finds all their characters at once. Write χ^λ for the character of V^λ (Theorem 12.6). A class function becomes a symmetric polynomial, its* Frobenius characteristic*; the irreducible characters become the* Schur functions *s_λ, quotients of two determinants; and one coefficient gives any value $\chi^\lambda(\mu)$.*

By the end of this lecture you will be able to:

- Prove that for $N \geq n$ the characteristic is an isometry $\mathrm{CF}(S_n) \to \Lambda^n$ that turns restriction to S_{n-1} ($n \geq 2$) into $\partial/\partial p_1$.
- Prove that for $N \geq n$ the s_λ, $\lambda \vdash n$, are an orthonormal basis of Λ^n, with $\partial s_\lambda/\partial p_1 = \sum_{\mu=\lambda-\square} s_\mu$ for $n \geq 2$.
- Prove that $\mathrm{ch}(\chi^\lambda) = s_\lambda$ for every $\lambda \vdash n$ when $N \geq n$.
- Compute any value $\chi^\lambda(\mu)$ by the Frobenius formula in $N \geq \ell(\lambda)$ variables.

13.1 Symmetric functions and the characteristic

Fix $n \geq 1$ and $N \geq n$ variables $x_1, \ldots, x_N$. For $\alpha \in \mathbb{Z}^N_{\geq 0}$, write $x^\alpha = x_1^{\alpha_1} \cdots x_N^{\alpha_N}$, and $[x^\alpha]f$ for the coefficient of x^α in a polynomial f. A partition λ of length at most N is padded with zeros to length N. Recall that $\ell(\lambda)$ is the number of its parts (Definition 11.13), and $m_i(\lambda)$ is the number of parts equal to i (Definition 2.15).

Definition 13.1 (Symmetric polynomials). Λ^n is the space of polynomials f in $x_1, \ldots, x_N$ with complex coefficients, homogeneous of degree n and *symmetric*: $f(x_{\sigma(1)}, \ldots, x_{\sigma(N)}) = f(x_1, \ldots, x_N)$ for every $\sigma \in S_N$. For $\lambda \vdash n$, the *monomial symmetric polynomial* m_λ is the sum of the distinct x^α with α a rearrangement of λ.

A symmetric f has the same coefficient at x^α as at its rearrangement x^λ, so

$$f = \sum_{\lambda \vdash n} [x^\lambda] f \; m_\lambda,$$

and the $p(n)$ monomial symmetric polynomials m_λ, $\lambda \vdash n$, form a basis of Λ^n, as $\ell(\lambda) \leq n \leq N$. Setting $x_{N+1} = 0$ sends each m_λ in $N+1$ variables to m_λ in N: an isomorphism, which also keeps the p_μ and h_μ below. Thus these polynomials, the Hall product and the characteristic defined below are unchanged by this identification in the range $N \geq n$.

Definition 13.2 (Power sums and complete sums). For $k \geq 1$,

$$p_k = x_1^k + \cdots + x_N^k,$$

and h_k is the sum of all monomials of degree k. For a partition $\mu = (\mu_1, \ldots, \mu_\ell)$, $p_\mu = p_{\mu_1} \cdots p_{\mu_\ell}$ and $h_\mu = h_{\mu_1} \cdots h_{\mu_\ell}$. For degree 0, the empty partition $\varnothing$ is the unique partition of 0. Set $h_0 = 1$ and use the empty product convention $p_\varnothing = h_\varnothing = 1$.

Subscripts keep p_k, h_k apart from $p(n)$ and group elements.

Lemma 13.3 (The power-sum basis). *The p_μ, $\mu \vdash n$, are a basis of Λ^n.*

Proof. Expanding $p_\mu = \prod_j (\sum_i x_i^{\mu_j})$, each monomial puts every part μ_j on one variable. Its exponents, sorted, are μ, or, if two parts share a variable, a partition of smaller length. And x^μ occurs $\prod_i m_i(\mu)!$ times, once for each way to send the parts to $x_1, \dots, x_\ell$ as x^μ needs:

$$p_\mu = \Big(\prod_i m_i(\mu)!\Big) m_\mu + (\text{terms } m_\lambda,\ \ell(\lambda) < \ell(\mu)).$$

List the partitions of n by decreasing length: the matrix taking the m_λ to the p_μ is triangular with non-zero diagonal. ∎

Definition 13.4 (The Hall product and the characteristic). The *Hall product* on Λ^n is

$$\langle p_\lambda, p_\mu \rangle = z_\lambda \, \delta_{\lambda\mu},$$

extended antilinearly in the first slot and linearly in the second, like $\langle \cdot, \cdot \rangle_{S_n}$. The *Frobenius characteristic* of $\chi \in \mathrm{CF}(S_n)$ is

$$\begin{aligned} \mathrm{ch}(\chi) &= \sum_{\mu \vdash n} \chi(\mu) \frac{p_\mu}{z_\mu} \\ &= \frac{1}{n!} \sum_{\sigma \in S_n} \chi(\sigma)\, p_{\mu(\sigma)}, \end{aligned}$$

where $\chi(\mu)$ is the value on the class of type μ, of size $n!/z_\mu$ (Theorem 2.16), and $\mu(\sigma)$ is the type of σ.

Differentiating $p_\mu = p_1^{m_1(\mu)} p_2^{m_2(\mu)} \cdots$ in p_1 defines a linear map $\partial/\partial p_1 \colon \Lambda^n \to \Lambda^{n-1}$ on this basis ($\Lambda^0 = \mathbb{C}$):

$$\frac{\partial p_\mu}{\partial p_1} = m_1(\mu)\, p_{\mu^-},$$

where μ^- is μ with one part 1 removed when $m_1(\mu) > 0$. The derivative is 0 when $m_1(\mu) = 0$.

Theorem 13.5 (The characteristic). *Let $n \geq 1$ and $N \geq n$.*

(i) $\mathrm{ch} \colon \mathrm{CF}(S_n) \to \Lambda^n$ *is a linear isomorphism, and* $\langle \mathrm{ch}\,\varphi, \mathrm{ch}\,\psi \rangle = \langle \varphi, \psi \rangle_{S_n}$.

(ii) For $\chi \in \mathrm{CF}(S_n)$ and $\mu \vdash n$,

$$[x^\mu]\,\mathrm{ch}(\chi) = \langle \chi_{M^\mu}, \chi \rangle_{S_n}.$$

So for a representation V, $\mathrm{ch}(\chi_V)$ has non-negative integer coefficients; and

$$\mathrm{ch}(1_{S_n}) = h_n = \sum_{\mu \vdash n} \frac{p_\mu}{z_\mu}.$$

(iii) If $n \geq 2$, then $\mathrm{ch}\big(\mathrm{Res}^{S_n}_{S_{n-1}} \chi\big) = \partial\,\mathrm{ch}(\chi)/\partial p_1$.

Proof. (i) ch takes the indicators of the classes, a basis of $\mathrm{CF}(S_n)$, to the basis p_μ/z_μ of Λ^n

(Lemma 13.3). And

$$\begin{aligned}\langle \operatorname{ch}\varphi, \operatorname{ch}\psi\rangle &= \sum_{\mu} \overline{\varphi(\mu)}\,\psi(\mu)\,\frac{z_\mu}{z_\mu^2}\\ &= \frac{1}{n!}\sum_{\mu}\frac{n!}{z_\mu}\,\overline{\varphi(\mu)}\,\psi(\mu)\\ &= \langle \varphi,\psi\rangle_{S_n},\end{aligned}$$

by the class form of $\langle\cdot,\cdot\rangle_{S_n}$ (Lecture 6).

(ii) *The count.* Let $\mu = (\mu_1,\dots,\mu_\ell) \vdash n$, with $\ell = \ell(\mu)$, and fix $\sigma \in S_n$. Recall from Proposition 8.18 that $M^\mu \cong \mathbb{C}^{X_\mu}$, where X_μ consists of ordered tuples of pairwise disjoint blocks A_i of sizes μ_i whose union is $\{1,\dots,n\}$. A tuple is fixed by σ exactly when every block is a union of cycles. Blocks of equal size remain distinguished by their positions.

Let $\nu = \mu(\sigma)$. Then

$$p_\nu = \prod_j \Big(\sum_i x_i^{\nu_j}\Big)$$

has one factor for each cycle of σ. A term assigns each cycle to a variable. It gives x^μ exactly when the cycles assigned to x_i cover μ_i points for every $i \le \ell$, and no cycle is assigned to a later variable. These cycles form a tuple in X_μ fixed by σ. Conversely, each fixed tuple assigns every cycle to the variable indexed by its block. Thus $[x^\mu]p_\nu$ counts the fixed tuples and equals $\chi_{M^\mu}(\sigma)$.

The average. As χ_{M^μ} is real,

$$\begin{aligned}[x^\mu]\operatorname{ch}(\chi) &= \frac{1}{n!}\sum_{\sigma}\chi(\sigma)\,\chi_{M^\mu}(\sigma)\\ &= \langle \chi_{M^\mu},\chi\rangle_{S_n}.\end{aligned}$$

For $\chi = \chi_V$ it is $\dim \operatorname{Hom}_{S_n}(M^\mu, V)$ (Theorem 6.7). Every degree-n exponent vector rearranges to a partition μ, and $\operatorname{ch}(\chi_V)$ is symmetric. So every coefficient is a non-negative integer.

The trivial character. For $\chi = 1_{S_n}$ it is the number of orbits on X_μ (Example 5.9). The action is transitive (Proposition 8.18), so there is one orbit. Every monomial of degree n has coefficient 1, as in h_n.

(iii) If $m_1(\mu) \ge 1$, then $z_\mu = m_1(\mu)\,z_{\mu^-}$, so

$$\frac{\partial}{\partial p_1}\frac{p_\mu}{z_\mu} = \frac{p_{\mu^-}}{z_{\mu^-}},$$

and the terms with $m_1(\mu) = 0$ vanish. Type ν in S_{n-1} is type $\nu \cup (1)$ in S_n, so

$$\begin{aligned}\frac{\partial \operatorname{ch}(\chi)}{\partial p_1} &= \sum_{\nu\vdash n-1}\chi\big(\nu\cup(1)\big)\frac{p_\nu}{z_\nu}\\ &= \operatorname{ch}\big(\operatorname{Res}^{S_n}_{S_{n-1}}\chi\big).\end{aligned}$$

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Example 13.6 (Guard: z_μ, not the class size)**.** For $n = 2$, $\operatorname{ch}(1_{S_2}) = (p_1^2 + p_2)/2$, and z_μ is 2 for both classes:

$$\big\langle \operatorname{ch}1_{S_2}, \operatorname{ch}1_{S_2}\big\rangle = \frac{1}{4}(2+2) = 1.$$

With the class sizes, 1 and 1, in place of the z_μ, it is

$$\frac{1}{4}(1+1) = \frac{1}{2}:$$

the isometry, and Cauchy's identity in § 13.2, need the z_μ.

13.2 Schur functions, Cauchy and $\partial/\partial p_1$

We construct an orthonormal basis of Schur functions with a one-box derivative rule. That rule matches restriction of characters and will let us identify the basis with the irreducible characters.

For $\alpha \in \mathbb{Z}^N_{\geq 0}$, the *alternant* is

$$a_\alpha = \det\left(x_j^{\alpha_i}\right)_{i,j=1}^N = \sum_{w \in S_N} \operatorname{sgn}(w) \prod_i x_{w(i)}^{\alpha_i},$$

and $\delta = (N-1, N-2, \ldots, 0)$, the *staircase* (a Kronecker $\delta_{\lambda\mu}$ carries two indices). Exchanging two variables exchanges two columns: a_α is *antisymmetric*, changing sign. If two α_i are equal, $a_\alpha = 0$; and permuting the entries of α by u multiplies a_α by $\operatorname{sgn}(u)$.

Lemma 13.7 (Alternants).

(i) An antisymmetric polynomial f in $x_1, \ldots, x_N$, with coefficients in a complex polynomial ring in other variables, is $\prod_{i<j}(x_i - x_j) \cdot g$ with g symmetric.

(ii) $a_\delta = \prod_{i<j}(x_i - x_j)$, of degree $N-1$ in each x_i.

Proof. (i) *One factor.* Dividing by $x_i - x_j$ as a polynomial in x_i leaves the remainder $f|_{x_i = x_j}$. Substituting $x_i = x_j$ gives the same result for f and for f with x_i, x_j exchanged. Antisymmetry makes these results negatives of each other. As the coefficient ring has characteristic zero, the remainder is 0.

All factors. Divide by the $x_i - x_j$ one at a time. Write f as the product of the earlier factors and the remaining quotient. After substituting $x_i = x_j$, the earlier distinct factors remain non-zero polynomials. Their product is non-zero in the polynomial integral domain, while the substituted f is 0 by the first step. So the substituted remaining quotient is 0. Thus the next factor divides that quotient. The product $\prod_{i<j}(x_i - x_j)$ is antisymmetric: an adjacent exchange negates one factor and permutes the others. Adjacent exchanges generate S_N, so the quotient g is symmetric.

(ii) By antisymmetric divisibility, a_δ is the product times a constant g, as both have degree $N(N-1)/2$. And $g = 1$, by induction on N: the coefficient of x_1^{N-1} is $a_\delta(x_2, \ldots, x_N)$ in the determinant (along the first column) and $\prod_{2 \leq i<j}(x_i - x_j)$ in the product. The product has degree $N-1$ in each variable, since each variable occurs in exactly $N-1$ factors. ∎

Definition 13.8 (Schur functions). For a partition λ with $\ell(\lambda) \leq N$, the *Schur function* is

$$s_\lambda = \frac{a_{\lambda+\delta}}{a_\delta},$$

a symmetric polynomial of degree $|\lambda|$, by Lemma 13.7.

For the empty partition, $s_\varnothing = a_\delta/a_\delta = 1$. The subscript of s_λ is a partition; a Coxeter generator s_i has an integer.

Example 13.9 (A Schur function). For $N = n = 2$, the quotient defining $s_{(2)}$ gives

$$s_{(2)} = \frac{x_1^3 - x_2^3}{x_1 - x_2} = x_1^2 + x_1x_2 + x_2^2 = h_2.$$

To define $h_n(xy)$, first form h_n in N^2 independent variables ξ_{ij}, and then substitute $\xi_{ij} = x_iy_j$. Different monomials in the ξ_{ij} can give the same monomial after substitution; their coefficients are retained.

Lemma 13.10 (Cauchy's identity). *In $x_1, \dots, x_N$ and $y_1, \dots, y_N$, for every $n \geq 0$,*

$$\sum_{\lambda \vdash n,\ \ell(\lambda) \leq N} a_{\lambda+\delta}(x)\, a_{\lambda+\delta}(y) = a_\delta(x)\, a_\delta(y)\, h_n(xy).$$

Proof. Work with formal power series (infinite sums of monomials), with $1/(1-u) = \sum_k u^k$.

The determinant. By Leibniz,

$$\begin{aligned} \det\Big(\frac{1}{1-x_i y_j}\Big) &= \sum_w \operatorname{sgn}(w) \prod_i \sum_{k_i \geq 0} (x_i y_{w(i)})^{k_i} \\ &= \sum_{k \in \mathbb{Z}_{\geq 0}^N} x^k\, a_k(y). \end{aligned}$$

Here $a_k(y) = 0$ unless the k_i are distinct. Each strictly decreasing non-negative N-tuple is uniquely $\lambda + \delta$ for a partition λ padded with zeros. The tuples k that rearrange one $\lambda + \delta$ collect, with their signs, to $a_{\lambda+\delta}(x)\, a_{\lambda+\delta}(y)$.

Cauchy's determinant. Let $P = \prod_{i,j}(1 - x_i y_j)$. Then

$$P \det\Big(\frac{1}{1-x_i y_j}\Big) = \sum_w \operatorname{sgn}(w) \prod_i \prod_{j \neq w(i)} (1 - x_i y_j)$$

is a polynomial, antisymmetric in the x_i and in the y_j, of degree at most $N-1$ in each variable. By Lemma 13.7(i) in x, and comparing degrees in each x_i, it is $a_\delta(x)\, g$ with g free of x; by antisymmetric divisibility in y, g is $a_\delta(y)$ times a polynomial. The degree bound $N-1$ in each y_j makes that polynomial a constant κ.

The constant. The geometric-series expansion gives $1/P$ as $\sum_{n \geq 0} h_n(xy)$, first expanding in the independent variables ξ_{ij} and then substituting $\xi_{ij} = x_i y_j$. Thus

$$\sum_\lambda a_{\lambda+\delta}(x)\, a_{\lambda+\delta}(y) = \kappa\, a_\delta(x)\, a_\delta(y) \sum_{n \geq 0} h_n(xy).$$

The terms of total degree $N(N-1)$ are the $\lambda = \varnothing$ term and $\kappa\, a_\delta(x)\, a_\delta(y)$, so $\kappa = 1$. The terms of total degree $N(N-1) + 2n$ give the lemma. ∎

Theorem 13.11 (Schur functions). *Let $n \geq 1$ and $N \geq n$.*

(i) The s_λ, $\lambda \vdash n$, are an orthonormal basis of Λ^n.

(ii) For $\mu \vdash n-1$, $p_1 s_\mu = \sum_{\lambda = \mu + \square} s_\lambda$; in particular $s_{(1)} = p_1$.

(iii) For $n \geq 2$ and $\lambda \vdash n$,

$$\frac{\partial s_\lambda}{\partial p_1} = \sum_{\mu = \lambda - \square} s_\mu.$$

Proof. (i) *Cauchy.* Divide Lemma 13.10 by $a_\delta(x)\, a_\delta(y)$, and apply Theorem 13.5(ii) in the $N^2 \geq n$ independent variables ξ_{ij}. After substituting $\xi_{ij} = x_i y_j$, the power sum becomes $p_k(xy) = p_k(x)\, p_k(y)$:

$$\begin{aligned} \sum_{\lambda \vdash n} s_\lambda(x)\, s_\lambda(y) &= h_n(xy) \\ &= \sum_{\rho \vdash n} \frac{p_\rho(x)\, p_\rho(y)}{z_\rho}. \end{aligned}$$

Coefficients. Write $s_\lambda = \sum_\rho b_{\lambda\rho}\, p_\rho$, with $b_{\lambda\rho}$ rational. Indeed the p_ρ form a basis and their transition matrix to the monomial basis has rational entries and a rational inverse; the

coefficients of s_λ are rational by polynomial division of its rational numerator by a_δ. The $p_\rho(x)\,p_\tau(y)$ are linearly independent, as bases in separate variables, so

$$\sum_\lambda b_{\lambda\rho}\,b_{\lambda\tau} = \frac{\delta_{\rho\tau}}{z_\rho}.$$

Both indices run over the partitions of n, so the real matrix with entries $b_{\lambda\rho}\sqrt{z_\rho}$ is square. It has orthonormal columns, hence orthonormal rows:

$$\langle s_\lambda, s_\mu\rangle = \sum_\rho b_{\lambda\rho}\,b_{\mu\rho}\,z_\rho = \delta_{\lambda\mu}.$$

There are $p(n)$ orthonormal functions in the $p(n)$-dimensional space Λ^n, so they form a basis.

(ii) Multiplying the term of w in a_α by $x_{w(i)}$ raises α_i by 1, so

$$p_1 a_\alpha = \sum_{i=1}^{N} a_{\alpha+\varepsilon_i},$$

with ε_i the i-th unit vector. Take $\alpha = \mu + \delta$. If $\mu+\varepsilon_i$ is a partition, it is $\mu+\square$, and each $\mu+\square$ arises so, as $\ell(\mu+\square) \le N$. Otherwise $\mu_{i-1} = \mu_i$, so $\alpha+\varepsilon_i$ has two equal entries, and $a_{\alpha+\varepsilon_i} = 0$. Divide by a_δ.

(iii) *Adjointness.* For $f \in \Lambda^{n-1}$ and $g \in \Lambda^n$,

$$\langle p_1 f, g\rangle = \langle f, \partial g/\partial p_1\rangle\,.$$

On $f = p_\nu$ and $g = p_\mu$, both sides vanish unless $\mu = \nu \cup (1)$, and then they are z_μ and $m_1(\mu)\,z_\nu$, equal by the proof of Theorem 13.5(iii).

Expansion. Apply Schur orthonormality, adjointness and the one-box rule:

$$\begin{aligned}\frac{\partial s_\lambda}{\partial p_1} &= \sum_{\mu\vdash n-1} \langle s_\mu, \partial s_\lambda/\partial p_1\rangle\; s_\mu \\ &= \sum_\mu \langle p_1 s_\mu, s_\lambda\rangle\; s_\mu \\ &= \sum_{\mu=\lambda-\square} s_\mu. \qquad \blacksquare\end{aligned}$$

13.3 Integrality and predecessors

We next prove integrality of the Schur functions' preimages. Together with the one-box rule and the predecessor lemma, this will identify those preimages with irreducible characters.

Lemma 13.12 (Extraction). *For $f \in \Lambda^n$ and $\lambda \vdash n$ ($N \ge n$),*

$$\langle s_\lambda, f\rangle = [x^{\lambda+\delta}](a_\delta f).$$

Proof. Expand in the orthonormal basis (Theorem 13.11):

$$f = \sum_\nu \langle s_\nu, f\rangle\; s_\nu,$$

$$a_\delta f = \sum_\nu \langle s_\nu, f\rangle\; a_{\nu+\delta}.$$

The exponents of $a_{\nu+\delta}$ are the rearrangements of $\nu+\delta$, of which only $\nu+\delta$, with coefficient 1, is strictly decreasing, as $\lambda+\delta$ is. So $[x^{\lambda+\delta}]a_{\nu+\delta}$ is 1 if $\nu = \lambda$, and 0 otherwise. $\blacksquare$

A *virtual character* of S_n is an integral combination of irreducible characters.

Corollary 13.13 (Integrality). *For $\lambda \vdash n$, $\mathrm{ch}^{-1}(s_\lambda)$ is a virtual character.*

Proof. Let $\varphi = \mathrm{ch}^{-1}(s_\lambda)$ and $U \in \mathrm{Irr}(S_n)$. By Theorem 13.5(i) and Lemma 13.12,

$$\langle \chi_U, \varphi \rangle_{S_n} = \langle \mathrm{ch}\,\chi_U, s_\lambda \rangle = \overline{[x^{\lambda+\delta}](a_\delta \,\mathrm{ch}\,\chi_U)},$$

an integer, as a_δ and $\mathrm{ch}\,\chi_U$ are integral (Theorem 13.5(ii)). The irreducible characters are an orthonormal basis of $\mathrm{CF}(S_n)$ (Theorem 7.11(ii)), so $\varphi = \sum_U \langle \chi_U, \varphi \rangle_{S_n} \chi_U$. ∎

Example 13.14 (Guard: norm one is not enough). Example 9.8's class function φ on S_3 has norm 1 and $\varphi(e) = 1$. But

$$\mathrm{ch}(\varphi) = \frac{1}{6}p_1^3 + \frac{1}{2}p_2p_1 - \frac{1}{3}p_3,$$

and

$$[x_1^3]\,\mathrm{ch}(\varphi) = \frac{1}{6} + \frac{1}{2} - \frac{1}{3} = \frac{1}{3},$$

which is $\langle 1_{S_3}, \varphi \rangle_{S_3}$: not an integer, so φ is not virtual (Theorem 13.5(ii)). Norm one and $\varphi(e) > 0$ make a character only with integrality (Example 6.11).

Lemma 13.15 (Predecessors). *For $n \geq 3$, a partition $\lambda \vdash n$ is determined by the set of partitions $\lambda - \square$. For $n = 2$, (2) and $(1,1)$ both have only (1).*

Proof. For $n = 2$, both partitions have only (1) as predecessor. Now let $n \geq 3$, and suppose $\lambda \neq \lambda'$ have the same predecessors, among them μ. Then $\lambda = \mu + \square$ and $\lambda' = \mu + \square'$ with $\square \neq \square'$, so the diagrams meet in μ, of $n - 1$ boxes.

Rectangles. Every predecessor of λ lies in both diagrams, so in μ, and is μ. Each distinct row length gives one removable box, at the end of the last row of that length. So one removable box means that all rows have the same length. Thus λ is a rectangle (a^b); likewise $\lambda' = (c^d)$.

The count. As $\lambda \neq \lambda'$, $a \neq c$; say $a < c$. Then $b > d$, as ab and cd are both n. The diagrams meet in (a^d), so

$$ad = n - 1 = ab - 1.$$

Hence $a(b - d) = 1$:

$$a = 1, \qquad b = d + 1.$$

Since $cd = n = ab = b$, we have $d(c - 1) = 1$:

$$d = 1, \qquad c = 2, \qquad n = 2.$$

This contradicts $n \geq 3$, so $\lambda = \lambda'$. ∎

13.4 The identification

We first use integrality, norm and the value at the identity to obtain an irreducible character. We then use restriction and predecessors to identify its label.

Theorem 13.16 (The characters of S_n). *For $n \geq 1$, $\lambda \vdash n$ and $N \geq n$,*

$$\mathrm{ch}(\chi^\lambda) = s_\lambda.$$

Proof. Let $\varphi_\lambda = \mathrm{ch}^{-1}(s_\lambda)$.

The value at e. Pairing $p_{(1^n)}$ with $s_\lambda = \sum_\mu \varphi_\lambda(\mu)\, p_\mu/z_\mu$ gives $\varphi_\lambda(e) = \langle p_1^n, s_\lambda\rangle$. The adjointness $\langle p_1 f, g\rangle = \langle f, \partial g/\partial p_1\rangle$ was proved in Theorem 13.11. Using it and the one-box derivative rule, for $n \geq 2$,

$$\begin{aligned}\langle p_1^n, s_\lambda\rangle &= \langle p_1^{n-1}, \partial s_\lambda/\partial p_1\rangle \\ &= \sum_{\mu=\lambda-\square} \langle p_1^{n-1}, s_\mu\rangle,\end{aligned}$$

and $\langle p_1, s_{(1)}\rangle = 1$. So $\varphi_\lambda(e)$ counts the paths from (1) to λ in Young's graph, by induction on n. There is at least one such path, from the row tableau, so $f^\lambda \geq 1$ (Definition 12.1).

Irreducible. By Theorem 13.5(i) and Theorem 13.11(i),

$$\langle \varphi_\lambda, \varphi_\mu\rangle_{S_n} = \langle s_\lambda, s_\mu\rangle = \delta_{\lambda\mu}.$$

So φ_λ is virtual (Corollary 13.13), of norm 1 and positive at e. Its integer coefficients in the irreducible character basis have squares summing to 1, so it is one irreducible character or its negative. The positive value at e chooses the positive sign (Example 6.11).

Counting. There are $p(n)$ distinct φ_λ and $p(n)$ irreducibles (Theorem 12.6), so $\varphi_\lambda = \chi^{\theta(\lambda)}$ for a bijection θ of partitions.

$n \leq 2$. $n = 1$ is clear. For $n = 2$, $\langle s_\lambda, p_2\rangle$ sums the coefficients of the $x^{\lambda+\delta-2\varepsilon_i}$ in a_δ (Lemma 13.12); only $i = 1$ gives a rearrangement of δ: subtracting 2 from a later entry leaves the first entry of $\lambda+\delta$ at least N, too large for a rearrangement of δ. For $i = 1$ the rearrangement is δ for (2), and δ with its first two entries exchanged for $(1,1)$. So the values $\overline{\varphi_\lambda((2))}$ are

$$\langle s_{(2)}, p_2\rangle = 1, \qquad \langle s_{(1,1)}, p_2\rangle = -1,$$

so

$$\varphi_{(2)} = 1_{S_2}, \qquad \varphi_{(1,1)} = \mathrm{sgn},$$

which are $\chi^{(2)}$ and $\chi^{(1,1)}$ (§ 12.2).

$n \geq 3$. With $\mathrm{ch}(\chi^\mu) = s_\mu$ for $\mu \vdash n-1$ (induction), Theorem 13.5(iii) and Theorem 13.11(iii) give

$$\begin{aligned}\mathrm{ch}\big(\mathrm{Res}^{S_n}_{S_{n-1}} \chi^{\theta(\lambda)}\big) &= \sum_{\mu=\lambda-\square} s_\mu \\ &= \mathrm{ch}\Big(\sum_{\mu=\lambda-\square} \chi^\mu\Big).\end{aligned}$$

The characteristic is injective (Theorem 13.5), so the displayed equality is an equality of characters. By branching (Theorem 12.6(iv)), the restricted character is also $\sum_{\nu=\theta(\lambda)-\square} \chi^\nu$. The irreducible characters for distinct partitions are distinct and linearly independent (Corollary 6.8(v)); each appears once in either sum. Thus λ and $\theta(\lambda)$ have the same predecessors, so $\theta(\lambda) = \lambda$ (Lemma 13.15). ∎

13.5 The Frobenius formula

Corollary 13.17 (The Frobenius formula). *For $\lambda, \mu \vdash n$, in any number $N \geq \ell(\lambda)$ of variables,*

$$\chi^\lambda(\mu) = [x^{\lambda+\delta}](a_\delta\, p_\mu).$$

For $N \geq n$, the formula is equivalently $p_\mu = \sum_\lambda \chi^\lambda(\mu)\, s_\lambda$.

Proof. $N \geq n$. By Lemma 13.12 and Theorem 13.16,

$$[x^{\lambda+\delta}](a_\delta\, p_\mu) = \langle \operatorname{ch}\chi^\lambda, p_\mu \rangle = \overline{\chi^\lambda(\mu)},$$

an integer, so real. The second form is the expansion of p_μ in the orthonormal basis s_λ.

Fewer variables. Let $\ell(\lambda) \leq N$. In $N+1$ variables the last entry of $\lambda+\delta$ is 0, so setting $x_{N+1} = 0$ keeps the coefficient. By Lemma 13.7(ii),

$$a_\delta(x_1, \dots, x_N, 0) = x_1 \cdots x_N\, a_\delta(x_1, \dots, x_N),$$

with the staircase written with N entries on the right; the factor $x_1 \cdots x_N$ lowers the first N entries of $\lambda+\delta$ by 1, to the $\lambda+\delta$ of N variables. So the coefficient does not change from $N+1$ to N, and the formula descends from $N \geq n$. ∎

13.6 Worked example

We compute characteristics of familiar S_3 representations and extract an entry of the S_4 character table.

$n = 3$. With χ_E from Example 6.4, and the fixed points of $M^{(2,1)} \cong \mathbb{C}^3$,

μ	(1^3)	$(2,1)$	(3)
z_μ	6	2	3
χ_E	2	0	−1
$\chi_{M^{(2,1)}}$	3	1	0

So

$$\operatorname{ch}(\chi_E) = \frac{2}{6}p_1^3 - \frac{1}{3}p_3 = \frac{1}{3}(p_1^3 - p_3).$$

As $E = V^{(2,1)}$, this is $s_{(2,1)}$. And

$$\operatorname{ch}(\chi_{M^{(2,1)}}) = \frac{1}{2}p_1^3 + \frac{1}{2}p_2p_1 = h_2h_1,$$

as, by Theorem 13.5(ii),

$$h_2 = \frac{1}{2}(p_1^2 + p_2), \qquad h_3 = \frac{1}{6}(p_1^3 + 3p_2p_1 + 2p_3).$$

So

$$h_2h_1 - h_3 = \frac{1}{6}(2p_1^3 - 2p_3) = \operatorname{ch}(\chi_E).$$

Injectivity gives $\chi_E = \chi_{M^{(2,1)}} - 1_{S_3}$, recovering Lecture 5's splitting.

An entry of S_4's table. Use $N = 2$ and $\lambda+\delta = (3,2)$. The Frobenius formula permits $N = 2$ because $\ell((2,2)) = 2$:

$$\begin{aligned} a_\delta\, p_{(3,1)} &= (x_1 - x_2)(x_1^3 + x_2^3)(x_1 + x_2) \\ &= x_1^5 - x_1^3x_2^2 + x_1^2x_2^3 - x_2^5, \end{aligned}$$

so $\chi^{(2,2)}((3,1)) = -1$, matching § 6.6's table, since $V^{(2,2)} \cong \rho_E \circ \pi$ (§ 12.4).

Self-check. Compute $\chi^{(3,1)}((2,2))$ by the Frobenius formula. *Answer:* $N = 2$ is allowed because $\ell((3,1)) = 2$. Then $\lambda+\delta = (4,1)$, and

$$\begin{aligned} (x_1 - x_2)(x_1^2 + x_2^2)^2 = x_1^5 - x_1^4x_2 + 2x_1^3x_2^2 \\ - 2x_1^2x_2^3 + x_1x_2^4 - x_2^5, \end{aligned}$$

so $\chi^{(3,1)}((2,2)) = -1$, as for χ_W on $(1\,2)(3\,4)$: $V^{(3,1)} \cong W$ (§ 12.4).

Summary.

- *The main results:* for $N \geq n$, ch is an isometry, with restriction $\partial/\partial p_1$ when $n \geq 2$ (Theorem 13.5); the s_λ are orthonormal (Theorem 13.11); $\mathrm{ch}(\chi^\lambda) = s_\lambda$ (Theorem 13.16); the Frobenius formula (Corollary 13.17).
- *Handed on:* $\mathrm{ch}(\chi^\lambda) = s_\lambda$ and $p_\mu = \sum_\lambda \chi^\lambda(\mu)\, s_\lambda$ for $N \geq n$ to Lecture 14.
- *Next:* Lecture 14 finds $s_\lambda(x_1, \ldots, x_d)$ as a character of GL_d.

References

Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed.: p_λ a basis, (2.12), and z_λ, p. 24; a_α and s_λ, (3.1), p. 40; Cauchy's identities, (4.1), p. 62, and (4.3), p. 63; dual bases and orthonormality, (4.6)–(4.8), pp. 63–64; Cauchy's determinant, §I.4 Ex. 6, pp. 67–68; the characteristic, (7.2)–(7.3), p. 113; $\mathrm{ch}(\chi^\lambda) = s_\lambda$ and the Frobenius formula, (7.6)–(7.8), p. 114.[1] Ceccherini-Silberstein, Scarabotti and Tolli, *Representation Theory of the Symmetric Groups*: Cauchy's determinant, Prop. 4.2.3, p. 173; Cauchy's identity for s_λ, Lemma 4.3.2, p. 186; the characteristic, §4.3.3, pp. 189–190. Fulton and Harris, *Representation Theory*: the Frobenius formula, 4.10, pp. 48–49.[2]

[1] Macdonald's products are bilinear, agreeing with ours on real functions.
[2] They index classes by numbers of j-cycles, and write Δ for a_δ and l for $\lambda + \delta$.